

Modeling and control of a nonlinear coupled spacecraft-fuel system

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Abstract

This paper considers the nonlinear interaction of the rigid body dynamics of a spacecraft with the sloshing dynamics of the liquid propellant in its partially filled spherical tank during an orbital transfer. Fuel slosh dynamics are described by a three-dimensional two-pendulum model characterizing the first two sloshing modes, and the coupled equations of motion of the vehicle and the fuel masses are derived by means of quasi-Lagrangian equations. In addition, this paper studies active nonlinear control methods to control the attitude of the spacecraft while suppressing the sloshing of fuel. Simulations are conducted to illustrate the effectiveness of these control methods.

Keywords: Fuel slosh modeling, Slosh control, Nonlinear controller, Sliding mode controller, Fuzzy controller, Lyapunov-based controller

1. Introduction

Attitude control is one of the most active research fields in aerospace engineering [1–6]. In particular, the attitude control problem for flexible spacecraft with fuel sloshing is the most critical problem of this area. In many space applications, it is necessary to place large and complex spacecraft (SC) in high al-

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titude orbits. Such spacecraft require a substantial fuel mass to total spacecraft mass ratio. For example, mass percentage of fuel for geosynchronous satellites is approximately 40 [7]. During translational or rotational accelerations of the SC, a large amount of fuel tend to move uncontrollably in the fuel tank, resulting in the sloshing phenomenon. The nonlinear coupling of the SC dynamics and the slosh dynamics can disturb the attitude of the SC from the desired position [8, 9]. Therefore, the fuel slosh and its effect on the attitude of space vehicles have been studied extensively during the last few decades [10–14]. The first failure due to fuel sloshing was in April 1957, when a Jupiter Intermediate Range Ballistic Missile (Jupiter IRBM) was terminated 93 seconds after its launch, and the more recent one was in 2007, when a circular coning oscillation occurred in Falcon One liquid-fuel launch vehicle 260 seconds after launch [15]. There are active and passive control methods for controlling the fuel slosh. The passive methods involve using baffles [16] and dividing the tank into smaller parts and some other methods exist, but these methods, in addition to increasing the weight, increase complexity and decrease performance of the SC [17]. Active fuel slosh control methods, which generally use throtttable lateral thrusters, gimballled thrusters, or reaction wheels, do not have the disadvantage of the passive methods. It must be noted that the increasing use of electric ion thrusters in spacecraft control systems might provide alternative solutions to the fuel sloshing problem. Numerous studies have been conducted in the field of fuel slosh and its effect on the attitude of the SC, proving that mechanical models [18] such as mass-spring [19, 20] and pendulum model [21–24] can approximate complicated fuel motion in the SC tank.

The main contribution of this paper is the derivation of a 3D multi-pendulum model of fuel slosh during thrust-vector maneuvering in space. 3D model is used for more realistic modeling. Two-pendulum model is used to characterize the first two slosh modes. Figure 1 shows these two modes. Figure 2 presents the schematic of the SC. The whole fuel consists of sloshing and still liquid parts; the still liquid fuel is modeled as a spherical mass rigidly attached to the tank. Figure 3 shows the schematic of the tank and fuel masses.

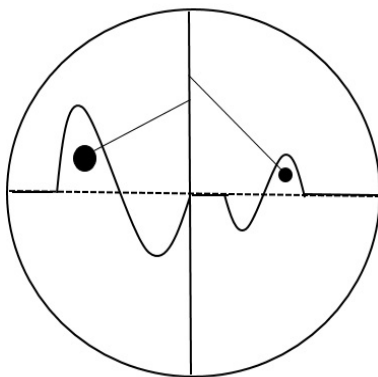


Figure 1: Two modes of fuel slosh in a spherical tank

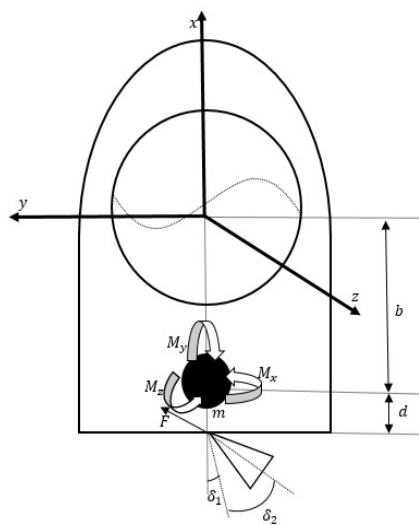


Figure 2: Model of the SC with a partially filled spherical fuel tank

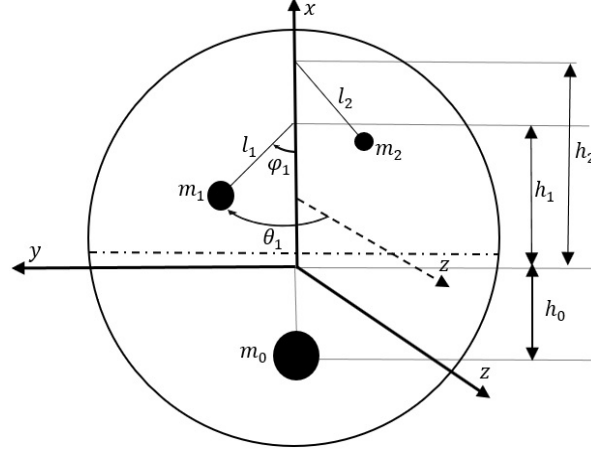


Figure 3: Model of sloshing and still liquid fuel in a spherical tank

The nonlinear equations of the coupled motion of the SC and pendulums are derived using a quasi-Lagrangian formulation [25]. In the literature, fuzzy, Lyapunov-based and sliding mode controllers [26–29] are designed to control the attitude of the SC while suppressing the fuel slosh. The SC has 6 degrees of freedom (DOF), and each pendulum has 2 DOF; thus, the complete system of the SC and pendulums has 10 DOF. As Fig. 2 shows, there are three moments acting at the center of mass of the SC as part of active control inputs, which are produced by reaction wheels and lateral thrusters, as well as two gimbal deflection angles. Therefore, the overall system has 5 control inputs to control 10 DOF, and thus it is underactuated.

2. Coupled dynamic equations

In this section, the coupled dynamic equations of the SC-fuel system will be formulated. For this aim, a SC maneuvering in 3D space from parking to working orbit is considered. Figure 2 shows the schematic of the SC. The SC is modeled by a rigid mass m with moments of inertia j_x , j_y and j_z with respect to the SC-fixed xyz -frame whose origin is at the tank center. The undisturbed fuel is described by a rigid spherical mass m_0 with the moment of inertia I_0 , and

the sloshing fuel is modeled by two-spherical pendulum masses ($m_i, I_i, i = 1, 2$) characterizing the first two modes of sloshing. Figure 3 shows the fuel model.

Equations of coupled motion of the SC-fuel system are derived using the quasi-Lagrange equations [25] given by

$$\frac{d}{dt} \frac{\partial L}{\partial \mathbf{V}} + \hat{\boldsymbol{\Omega}} \frac{\partial L}{\partial \mathbf{V}} = \boldsymbol{\tau}_t \quad (1)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \boldsymbol{\Omega}} + \hat{\boldsymbol{\Omega}} \frac{\partial L}{\partial \boldsymbol{\Omega}} + \hat{\mathbf{V}} \frac{\partial L}{\partial \mathbf{V}} = \boldsymbol{\tau}_r \quad (2)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\boldsymbol{\zeta}}} - \frac{\partial L}{\partial \boldsymbol{\zeta}} + \frac{\partial R}{\partial \dot{\boldsymbol{\zeta}}} = 0 \quad (3)$$

where $\mathbf{V}$ is velocity vector of the center of the tank, $\boldsymbol{\tau}_t$ and $\boldsymbol{\tau}_r$ are the vectors of generalized forces and moments acting on the SC, $\boldsymbol{\zeta}$ denotes the attitude angle vectors of the pendulums, which can be expressed as

$$\mathbf{V} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} + \hat{\boldsymbol{\Omega}} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (4)$$

$$\boldsymbol{\tau}_t = \begin{bmatrix} F \cos \delta_1 \cos \delta_2 \\ F \sin \delta_2 \\ -F \sin \delta_1 \cos \delta_2 \end{bmatrix}, \quad \boldsymbol{\tau}_r = \begin{bmatrix} M_x \\ M_y - F(b+d) \sin \delta_1 \cos \delta_2 \\ M_z - F(b+d) \sin \delta_2 \end{bmatrix}, \quad \boldsymbol{\zeta} = \begin{bmatrix} \varphi_i \\ 0 \\ \theta_i \end{bmatrix} \quad (5)$$

and R is the Rayleigh dissipation function defined as

$$R = \frac{1}{2} \sum_{i=1}^2 \epsilon_i (\dot{\varphi}_i^2 + \dot{\theta}_i^2) \quad (6)$$

Here, F is the main thruster force, δ_1 and δ_2 are the gimbal deflection angles about y and z axes, M_x , M_y and M_z denote the control moments about the center of mass of the SC, ϵ_i is the damping coefficient of i -th pendulum, d is the distance between the SC center of mass and the pivot point of the gimbal.

It is assumed that the altitude and the main thrust of the SC are large enough so that gravitational potential can be ignored. Thus, the Lagrangian

equals the kinetic energy, which can be written as

$$L = \frac{1}{2}m \left\| \mathbf{V}_c \right\|^2 + \frac{1}{2}m_0 \left\| \mathbf{V}_0 \right\|^2 + \frac{1}{2}\boldsymbol{\Omega}^T(\mathbf{J} + I_0\mathbf{I})\boldsymbol{\Omega} + \frac{1}{2}\sum_{i=1}^2[m_i \left\| \mathbf{V}_i \right\|^2 + (\boldsymbol{\omega}_i + \boldsymbol{\Omega})^T(I_i\mathbf{I})(\boldsymbol{\omega}_i + \boldsymbol{\Omega})] \quad (7)$$

where $\mathbf{V}_c$, $\mathbf{V}_0$, and $\mathbf{V}_i$ denote the velocity vectors of m , m_0 , and m_i , respectively, and $\boldsymbol{\Omega}$ and $\boldsymbol{\omega}_i$ are angular velocity vectors of m and m_i , respectively.

$$\mathbf{V}_c = \dot{\mathbf{r}}_c + \hat{\boldsymbol{\Omega}}\mathbf{r}_c, \quad \mathbf{V}_0 = \dot{\mathbf{r}}_0 + \hat{\boldsymbol{\Omega}}\mathbf{r}_0, \quad \mathbf{V}_i = \dot{\mathbf{r}}_i + \hat{\boldsymbol{\Omega}}\mathbf{r}_i \quad (8)$$

$$\boldsymbol{\Omega} = \begin{bmatrix} \Omega_1 \\ \Omega_2 \\ \Omega_3 \end{bmatrix}, \quad \boldsymbol{\omega}_i = \begin{bmatrix} -\dot{\theta}_i \\ \dot{\varphi}_i \cos \theta_i \\ -\dot{\varphi}_i \sin \theta_i \end{bmatrix} \quad (9)$$

In (7), $\mathbf{I}$ is 3×3 identity matrix, and $\mathbf{J}$ is the diagonal matrix of the SC moments of inertia, i.e.,

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{J} = \begin{bmatrix} j_x & 0 & 0 \\ 0 & j_y & 0 \\ 0 & 0 & j_z \end{bmatrix}$$

Attitude angles of the SC ($\vartheta = [\vartheta_1 \quad \vartheta_2 \quad \vartheta_3]^T$) are measured with respect a fixed reference frame. Accordingly, angular velocities of the SC are expressed by 321 angle sequences, but we assume that the attitude angles of the SC are small enough so that $\boldsymbol{\Omega}$ can be expressed as

$$\boldsymbol{\Omega} = \begin{bmatrix} 1 & 0 & -\sin \vartheta_2 \\ 0 & \cos \vartheta_2 & \sin \vartheta_1 \cos \vartheta_2 \\ 0 & -\sin \vartheta_1 & \cos \vartheta_1 \cos \vartheta_2 \end{bmatrix} \begin{bmatrix} \dot{\vartheta}_1 \\ \dot{\vartheta}_2 \\ \dot{\vartheta}_3 \end{bmatrix} \simeq \begin{bmatrix} \dot{\vartheta}_1 \\ \dot{\vartheta}_2 \\ \dot{\vartheta}_3 \end{bmatrix} \quad (10)$$

In (8) $\mathbf{r}_c$, $\mathbf{r}_0$ and $\mathbf{r}_i$ denote the position vectors of m , m_0 and m_i , respectively.

$$\mathbf{r}_c = \begin{bmatrix} x - b \\ y \\ z \end{bmatrix}, \quad \mathbf{r}_0 = \begin{bmatrix} x - h_0 \\ y \\ z \end{bmatrix}, \quad \mathbf{r}_i = \begin{bmatrix} x + h_i - l_i \cos \varphi_i \\ y + l_i \sin \varphi_i \sin \theta_i \\ z + l_i \sin \varphi_i \cos \theta_i \end{bmatrix} \quad (11)$$

where b is the distance between the tank center (origin) and the center of mass of the SC, h_0 and h_i are the distances of m_0 and the pivot point of i -th pendulum

from the origin, respectively; l_i is the length of i -th pendulum, and φ_i and θ_i are the attitude angles of the i -th pendulum.

By applying (1)-(3), equations of motion for the SC-fuel system are derived as follows

$$\begin{aligned}
(m + m_0)a_x + (mb + m_0h_0)(\Omega_2^2 + \Omega_3^2) + \sum_{i=1}^2 [m_i(a_x + l_i(\ddot{\varphi}_i \sin \varphi_i \\
+ \dot{\varphi}_i^2 \cos \varphi_i + 2\dot{\varphi}_i\Omega_2 \cos \varphi_i \cos \theta_i - 2\dot{\theta}_i\Omega_2 \sin \varphi_i \sin \theta_i - 2\dot{\varphi}_i\Omega_3 \cos \varphi_i \\
\sin \theta_i - 2\dot{\theta}_i\Omega_3 \sin \varphi_i \cos \theta_i + \dot{\Omega}_2 \sin \varphi_i \cos \theta_i - \dot{\Omega}_3 \sin \varphi_i \sin \theta_i + (\Omega_2^2 - \\
\Omega_3^2)(h_i - l_i \cos \varphi_i) + \Omega_1\Omega_3 \sin \varphi_i \cos \theta_i + \Omega_1\Omega_2 \sin \varphi_i \sin \theta_i))] = F \cos \delta_1 \\
\cos \delta_2
\end{aligned} \tag{12}$$

$$\begin{aligned}
(m + m_0)a_y - (mb + m_0h_0)(\dot{\Omega}_3 + \Omega_1\Omega_2) + \sum_{i=1}^2 [m_i(a_y + l_i(\ddot{\varphi}_i \cos \varphi_i \\
\sin \theta_i + \ddot{\theta}_i \sin \varphi_i \cos \theta_i - \dot{\varphi}_i^2 \sin \varphi_i \sin \theta_i - \dot{\theta}_i^2 \sin \varphi_i \sin \theta_i + 2\dot{\varphi}_i\Omega_1 \cos \varphi_i \\
\cos \theta_i + 2\dot{\theta}_i\Omega_1 \sin \varphi_i \sin \theta_i + 2\dot{\varphi}_i\Omega_3 \sin \varphi_i + 2\dot{\varphi}_i\dot{\theta}_i \cos \varphi_i \cos \theta_i - (\dot{\Omega}_1^2 - \\
\Omega_2\Omega_3) \sin \varphi_i \cos \theta_i - (\Omega_1^2 + \Omega_3^2) \sin \varphi_i \sin \theta_i + (\dot{\Omega}_3 + \Omega_1\Omega_2)(h_i - l_i \\
\cos \varphi_i))] = F \sin \delta_2
\end{aligned} \tag{13}$$

$$\begin{aligned}
(m + m_0)a_z + (mb + m_0h_0)(\dot{\Omega}_2 - \Omega_1\Omega_3) + \sum_{i=1}^2 [m_i(a_z + l_i(\ddot{\varphi}_i \cos \varphi_i \\
\cos \theta_i - \ddot{\theta}_i \sin \varphi_i \sin \theta_i - \dot{\varphi}_i^2 \sin \varphi_i \cos \theta_i - \dot{\theta}_i^2 \sin \varphi_i \cos \theta_i + 2\dot{\varphi}_i\Omega_1 \cos \varphi_i \\
\sin \theta_i + 2\dot{\theta}_i\Omega_1 \sin \varphi_1 \cos \theta_i - 2\dot{\varphi}_i\Omega_2 \sin \varphi_i - 2\dot{\varphi}_i\dot{\theta}_i \cos \varphi_i \sin \theta_i + (\dot{\Omega}_1 + \\
\Omega_2\Omega_3) \sin \varphi_i \sin \theta_i - (\Omega_1^2 + \Omega_2^2) \sin \varphi_i \sin \theta_i - (\dot{\Omega}_2^2 - \Omega_1\Omega_3)(h_i - l_i \\
\cos \varphi_i))] = -F \sin \delta_1 \cos \delta_2
\end{aligned} \tag{14}$$

$$\begin{aligned}
\dot{\Omega}_1(j_x + I_0) + \Omega_2\Omega_3(j_z - j_y) + I_i(\dot{\Omega}_1 - \ddot{\theta}_i) - I_i\Omega_3(\Omega_2 + \dot{\varphi}_i \cos \theta_i) + \\
I_i\Omega_2(\Omega_3 - \dot{\varphi}_i \sin \theta_i) + \sum_{i=1}^2 m_i l_i (\sin \varphi_i \cos \theta_i (-\dot{\Omega}_3 h_i - a_y) + \sin \varphi_i \sin \theta_i \\
(-\dot{\Omega}_2 h_i - a_z) - \Omega_3^2 h_i \cos \varphi_i - \Omega_2\Omega_3 h_i \cos \varphi_i + \Omega_1\Omega_3 h_i \sin \varphi_i \sin \theta_i + \\
\Omega_1\Omega_2 h_i \sin \varphi_i \cos \theta_i) = M_x
\end{aligned} \tag{15}$$

$$\begin{aligned}
& \dot{\Omega}_2(j_y + I_0) + \Omega_1\Omega_3(j_x - j_z) + I_i(\ddot{\varphi}_i \cos \theta_i - \dot{\varphi}_i \dot{\theta}_i \sin \theta_i + \dot{\Omega}_2) + \\
& I_i\Omega_3(\Omega_1 - \dot{\theta}_i) - I_i\Omega_1(\Omega_3 - \dot{\varphi}_i \sin \theta_i) + a_z(mb + m_0h_0) + (mb^2 + \\
& m_0h_0^2)(\dot{\Omega}_2 - \Omega_1\Omega_3) + \sum_{i=1}^2 [m_i(l_i(\sin \varphi_i \cos \theta_i a_x - \Omega_3^2 h_i \sin \varphi_i \cos \theta_i - \\
& \Omega_2\Omega_3 h_i \sin \varphi_i \cos \theta_i + \Omega_1\Omega_3 h_i \cos \varphi_i - \Omega_1\Omega_3 l_i \cos \varphi_i^2 + \cos \varphi_i(a_z - \\
& \dot{\Omega}_2 h_i + \dot{\Omega}_2 l_i \cos \varphi_i - 2\Omega_2 \dot{\varphi}_i l_i \sin \varphi_i)) - h_i(a_z - \dot{\Omega}_2 h_i + \dot{\Omega}_2 l_i \cos \varphi_i - \\
& 2\Omega_2 \dot{\varphi}_i l_i \sin \varphi_i + \dot{\Omega}_1 l_i \sin \varphi_i \sin \theta_i + 2\Omega_1 \dot{\varphi}_i l_i \cos \varphi_i \sin \theta_i + 2\Omega_1 \dot{\theta}_i l_i \sin \varphi_i \\
& \cos \theta_i + \ddot{\varphi}_i l_i \cos \varphi_i \cos \theta_i - \dot{\varphi}_i^2 l_i \sin \varphi_i \cos \theta_i - 2\dot{\varphi}_i \dot{\theta}_i l_i \cos \varphi_i \sin \theta_i - \ddot{\theta}_i \\
& l_i \sin \varphi_i \sin \theta_i - \dot{\theta}_i^2 l_i \sin \varphi_i \cos \theta_i) - h_i\Omega_1(\Omega_3 h_i - \Omega_3 l_i \cos \varphi_i - \Omega_1 l_i \sin \varphi_i \\
& \cos \theta_i)] = M_y - F(b + d) \sin \delta_1 \cos \delta_2
\end{aligned} \tag{16}$$

$$\begin{aligned}
& \dot{\Omega}_3(j_z + I_0) + \Omega_1\Omega_2(j_y - j_x) + I_i(\dot{\Omega}_3 - \ddot{\varphi}_i \sin \theta_i - \dot{\varphi}_i \dot{\theta}_i \cos \theta_i) - I_i\Omega_2 \\
& (\Omega_1 - \dot{\theta}_i) + I_i\Omega_2(\Omega_2 + \dot{\varphi}_i \cos \theta_i) - a_y(mb + m_0h_0) + (mb^2 + m_0h_0^2) \\
& (\dot{\Omega}_3 + \Omega_1\Omega_2) + \sum_{i=1}^2 [m_i(l_i(-a_x \sin \varphi_i \sin \theta_i - \cos \varphi_i(a_y - \dot{\Omega}_3 l_i \cos \varphi_i + \\
& 2\Omega_3 \dot{\varphi}_i l_i \sin \varphi_i) + \Omega_2\Omega_3 h_i \sin \varphi_i \cos \theta_i + \Omega_2^2 h_i \sin \varphi_i \sin \theta_i + \Omega_1\Omega_2 l_i \\
& \cos \varphi_i^2) + h_i(a_y + \dot{\Omega}_3 h_i - 2\dot{\Omega}_3 l_i \cos \varphi_i + 2\Omega_3 \dot{\varphi}_i l_i \sin \varphi_i - \dot{\Omega}_1 l_i \sin \varphi_i \\
& \cos \theta_i - \Omega_1 \dot{\varphi}_i l_i \cos \varphi_i \cos \theta_i + 2\Omega_1 \dot{\theta}_i l_i \sin \varphi_i \sin \theta_i + \ddot{\varphi}_i l_i \cos \varphi_i \sin \theta_i - \\
& \dot{\varphi}_i^2 l_i \sin \varphi_i \sin \theta_i + 2\dot{\varphi}_i \dot{\theta}_i l_i \cos \varphi_i \cos \theta_i + \ddot{\theta}_i l_i \sin \varphi_i \cos \theta_i - \dot{\theta}_i^2 l_i \sin \varphi_i \\
& \sin \theta_i) - h_i\Omega_1(-\Omega_2 h_i + 2\Omega_2 l_i \cos \varphi_i + \Omega_1 l_i \sin \varphi_i \sin \theta_i + \dot{\varphi}_i l_i \cos \varphi_i \\
& \cos \theta_i)] = M_z - F(b + d) \sin \delta_2
\end{aligned} \tag{17}$$

$$\begin{aligned}
& I_i(\cos \theta_i(\ddot{\varphi}_i \cos \theta_i + \dot{\Omega}_2) - \sin \theta_i(\dot{\Omega}_3 - \dot{\varphi}_i \sin \theta_i) - \dot{\theta}_i(\Omega_2 \sin \theta_i + \Omega_3 \\
& \cos \theta_i)) + m_i l_i(a_x \sin \varphi_i + \cos \varphi_i \sin \theta_i(a_y + \dot{\Omega}_3 h_i) + \cos \varphi_i \cos \theta_i(a_z - \\
& \dot{\Omega}_2 h_i) - \Omega_3 \sin \varphi_i(\Omega_3 h_i - \Omega_3 l_i \cos \varphi_i) + \Omega_2 \sin \varphi_i(-\Omega_2 h_i + \Omega_2 l_i \cos \varphi_i) \\
& + \Omega_1\Omega_3 h_i \cos \varphi_i \cos \theta_i + \Omega_1\Omega_2 h_i \cos \varphi_i \sin \theta_i) = 0, \quad i = 1, 2
\end{aligned} \tag{18}$$

$$\begin{aligned}
& I_i(\ddot{\theta}_i - \dot{\Omega}_1 + \dot{\varphi}_i \Omega_2 \sin \theta_i + \dot{\varphi}_i \Omega_3 \cos \theta_i) + m_i l_i (\sin \varphi_i \cos \theta_i (a_y + \dot{\Omega}_3 h_i) \\
& - \sin \varphi_i \sin \theta_i (a_z - \dot{\Omega}_2 h_i) - \Omega_1 \Omega_3 h_i \sin \varphi_i \sin \theta_i + \Omega_1 \Omega_2 h_i \sin \varphi_i \cos \theta_i) \quad (19) \\
& = 0, \quad i = 1, 2
\end{aligned}$$

Here $\mathbf{a} = [a_x \quad a_y \quad a_z]^T$ is the acceleration vector of the center of the tank, which can be computed as

$$\mathbf{a} = \dot{\mathbf{V}} + \hat{\boldsymbol{\Omega}} \mathbf{V} \quad (20)$$

3. Control law

In this section, three different control laws are designed for the control of the SC attitude while suppressing the sloshing modes. In particular, we design sliding mode, fuzzy and Lyapunov-based controllers. Now following [26], if we assume that there are no control inputs ($M_x = M_y = M_z = \delta_1 = \delta_2 = 0$) and the main thrust F is constant and positive, then the first equation of motion representing the motion of the SC along the longitudinal x axis becomes:

$$\begin{aligned}
& (m + m_0)a_x + (mb + m_0 h_0)(\Omega_2^2 + \Omega_3^2) + \sum_{i=1}^2 [m_i(a_x + l_i(\ddot{\varphi}_i \sin \varphi_i \\
& + \dot{\varphi}_i^2 \cos \varphi_i + 2\dot{\varphi}_i \Omega_2 \cos \varphi_i \cos \theta_i - 2\dot{\theta}_i \Omega_2 \sin \varphi_i \sin \theta_i - 2\dot{\varphi}_i \Omega_3 \cos \varphi_i \quad (21) \\
& \sin \theta_i - 2\dot{\theta}_i \Omega_3 \sin \varphi_i \cos \theta_i + \dot{\Omega}_2 \sin \varphi_i \cos \theta_i - \dot{\Omega}_3 \sin \varphi_i \sin \theta_i + (\Omega_2^2 - \\
& \Omega_3^2)(h_i - l_i \cos \varphi_i) + \Omega_1 \Omega_3 \sin \varphi_i \cos \theta_i + \Omega_1 \Omega_2 \sin \varphi_i \sin \theta_i))] = F
\end{aligned}$$

Consequently, the SC and slosh dynamics have a relative equilibrium given by

$$v_y = \tilde{v}_y, \quad v_z = \tilde{v}_z, \quad \vartheta = \tilde{\vartheta}, \quad \Omega = 0, \quad \zeta = \dot{\zeta} = 0 \quad (22)$$

If we assume $\tilde{v}_y$, $\tilde{v}_z$ and $\tilde{\vartheta}$ are zero, then (21) yields

$$a_x = \frac{F}{m + m_0 + \sum_{i=1}^2 m_i} \quad (23)$$

Now by transforming the original control inputs to new control inputs (u_{1-5}),

equations of coupled motion can be rewritten in a reduced form as

$$\dot{v}_y = u_1 - v_x \Omega_3 + v_z \Omega_1 \quad (24)$$

$$\dot{v}_z = u_2 - v_y \Omega_1 + v_x \Omega_1 \quad (25)$$

$$\dot{\Omega}_1 = u_3 \quad (26)$$

$$\dot{\Omega}_2 = u_4 \quad (27)$$

$$\dot{\Omega}_3 = u_5 \quad (28)$$

$$\begin{aligned} \ddot{\varphi}_i = & \frac{-1}{I_i \cos \theta_i^2} (I_i (\cos \theta_i (\dot{\Omega}_2) - \sin \theta_i (\dot{\Omega}_3 - \dot{\varphi}_i \sin \theta_i) - \dot{\theta}_i (\Omega_2 \sin \theta_i + \Omega_3 \\ & \cos \theta_i)) + m_i l_i (a_x \sin \varphi_i + \cos \varphi_i \sin \theta_i (a_y + \dot{\Omega}_3 h_i) + \cos \varphi_i \cos \theta_i (a_z \\ & - \dot{\Omega}_2 h_i) - \Omega_3 \sin \varphi_i (\Omega_3 h_i - \Omega_3 l_i \cos \varphi_i) + \Omega_2 \sin \varphi_i (-\Omega_2 h_i + \Omega_2 l_i \\ & \cos \varphi_i) + \Omega_1 \Omega_3 h_i \cos \varphi_i \cos \theta_i + \Omega_1 \Omega_2 h_i \cos \varphi_i \sin \theta_i)), \quad i = 1, 2 \end{aligned} \quad (29)$$

$$\begin{aligned} \ddot{\theta}_i = & \frac{-1}{I_i} (I_i (-\dot{\Omega}_1 + \dot{\varphi}_i \Omega_2 \sin \theta_i + \dot{\varphi}_i \Omega_3 \cos \theta_i) + m_i l_i (\sin \varphi_i \cos \theta_i (a_y \\ & + \dot{\Omega}_3 h_i) - \sin \varphi_i \sin \theta_i (a_z - \dot{\Omega}_2 h_i) - \Omega_1 \Omega_3 h_i \sin \varphi_i \sin \theta_i + \Omega_1 \Omega_2 h_i \\ & \sin \varphi_i \cos \theta_i)), \quad i = 1, 2 \end{aligned} \quad (30)$$

3.1. Sliding mode controller

In this section, we consider the reduced form of equations and define the following five sliding mode variables S_{1-5} :

$$S_1 = \left(\frac{d}{dt} + \lambda_1 \right)^{r_1} \int_0^t e_1(\tau) d\tau \quad (31)$$

$$S_2 = \left(\frac{d}{dt} + \lambda_2 \right)^{r_2} \int_0^t e_2(\tau) d\tau \quad (32)$$

$$S_3 = \left(\frac{d}{dt} + \lambda_3 \right)^{r_3} \int_0^t e_3(\tau) d\tau \quad (33)$$

$$S_4 = \left(\frac{d}{dt} + \lambda_4 \right)^{r_4} \int_0^t e_4(\tau) d\tau \quad (34)$$

$$S_5 = \left(\frac{d}{dt} + \lambda_5 \right)^{r_5} \int_0^t e_5(\tau) d\tau \quad (35)$$

Here $e_1 = v_y - v_{yd}$, $e_2 = v_z - v_{zd}$, $e_3 = \vartheta_1 - \vartheta_{1d}$, $e_4 = \vartheta_2 - \vartheta_{2d}$, $e_5 = \vartheta_3 - \vartheta_{3d}$, where v_{yd} , v_{zd} , ϑ_{1d} , ϑ_{2d} and ϑ_{3d} are the desired values of the velocities v_y and v_z and the attitude angles of the SC; r_{1-5} are the corresponding relative degrees ($r_{1-2} = 1$, $r_{3-5} = 2$), and λ_{1-5} are the positive constants defining the bandwidths of the error dynamics. The time derivatives of the sliding variables can be written as:

$$\dot{S}_1 = u_1 - v_x \Omega_3 + v_z \Omega_1 + \lambda_1 v_y \quad (36)$$

$$\dot{S}_2 = u_2 - v_y \Omega_1 + v_x \Omega_2 + \lambda_2 v_z \quad (37)$$

$$\dot{S}_3 = u_3 + 2\lambda_3 \vartheta_1 + \lambda_3^2 \Omega_1 \quad (38)$$

$$\dot{S}_4 = u_4 + 2\lambda_4 \vartheta_2 + \lambda_4^2 \Omega_2 \quad (39)$$

$$\dot{S}_5 = u_5 + 2\lambda_5 \vartheta_3 + \lambda_5^2 \Omega_3 \quad (40)$$

We derive u_{1-5} such that the following sliding conditions are satisfied

$$\frac{1}{2} \frac{dS_{1-2}}{dt} \leq -K_{1-2} |S_{1-2}| \quad (41)$$

$$\frac{1}{2} \frac{dS_{3-5}^2}{dt} \leq -K_{3-5} |S_{3-5}| \quad (42)$$

where K_{1-5} are positive constants. Therefore, the new control inputs (u_{1-5}) are obtained as:

$$u_1 = v_x \Omega_3 - v_z \Omega_1 - \lambda_1 v_y - K_1 \operatorname{sgn}(S_1) \quad (43)$$

$$u_2 = v_y \Omega_1 - v_x \Omega_2 - \lambda_2 v_z - K_2 \operatorname{sgn}(S_2) \quad (44)$$

$$u_3 = -2\lambda_3 \Omega_1 - \lambda_3^2 \vartheta_1 - K_3 \operatorname{sgn}(S_3) \quad (45)$$

$$u_4 = -2\lambda_4 \Omega_2 - \lambda_4^2 \vartheta_2 - K_4 \operatorname{sgn}(S_4) \quad (46)$$

$$u_5 = -2\lambda_5 \Omega_3 - \lambda_5^2 \vartheta_3 - K_5 \operatorname{sgn}(S_5) \quad (47)$$

where sgn is the signum function. In order to avoid the undesirable chattering, we use the continuous saturation function ($\operatorname{sat}(S_{1-5})$) instead of the signum function.

$$\operatorname{sat}(S_{1-5}) = \begin{cases} -1 : & \frac{S_{1-5}}{B_{1-5}} \leq -1 \\ \frac{S_{1-5}}{B_{1-5}} : & -1 < \frac{S_{1-5}}{B_{1-5}} \leq 1 \\ 1 : & \frac{S_{1-5}}{B_{1-5}} > 1 \end{cases} \quad (48)$$

Here B_{1-5} are the thin boundary layer widths.

3.2. Fuzzy controller

In this section, we design a fuzzy control law by expressing the control inputs as

$$u_1 = K_{p_1} v_y \quad (49)$$

$$u_2 = K_{p_2} v_z \quad (50)$$

$$u_3 = K_{p_3} \vartheta_1 + K_{d_3} \Omega_1 \quad (51)$$

$$u_4 = K_{p_4} \vartheta_2 + K_{d_4} \Omega_2 \quad (52)$$

$$u_5 = K_{p_5} \vartheta_3 + K_{d_5} \Omega_3 \quad (53)$$

where K_{p_q} and K_{d_s} ($q = 1 - 5$ and $s = 3 - 5$) are the controller gains obtained by the following equations:

$$K_{p_q} = K_{p_{q_{min}}} + (K_{p_{q_{max}}} - K_{p_{q_{min}}}) \bar{K}_{p_q} \quad (54)$$

$$K_{d_s} = K_{d_{s_{min}}} + (K_{d_{s_{max}}} - K_{d_{s_{min}}}) \bar{K}_{d_s} \quad (55)$$

Here the max and min indices denote the maximum and minimum values of the controller gains. Furthermore, $\bar{K}_{p_q}$ and $\bar{K}_{d_s}$ are the fuzzy output coefficient gains given by Mamdani logic. Fig. 4 indicates that there are two Gaussian membership functions (B and S) constituent for each output in $[0 \ 1]$. Also, initial data for Mamdani logic are v_y , v_z , ϑ_1 , ϑ_2 , ϑ_3 , Ω_1 , Ω_2 and Ω_3 . As Fig. 5 depicts, each initial data has five triangles (NM, NS, ZO, PS and PM) and two trapezoidal (NB and PB) membership functions. In this figure, e is the error of a fuzzy logical input and $\dot{e}$ is its derivative. There are quite a few rules for determining $\bar{K}_{p_{3-5}}$ and $\bar{K}_{d_{3-5}}$, which are defined using Table 1 and Table 2, respectively [30]. Moreover, if v_y gets its largest value (NB or PB), $\bar{K}_{p_1}$ is B, else $\bar{K}_{p_1}$ is S. Rules between $\bar{K}_{p_2}$ and v_z are defined the same way as the rules between $\bar{K}_{p_1}$ and v_y .

3.3. Lyapunov-based controller

For designing a nonlinear Lyapunov-based controller, we first define a candidate Lyapunov function $\mathcal{V}$. Consider the reduced equations (24)-(30) and define

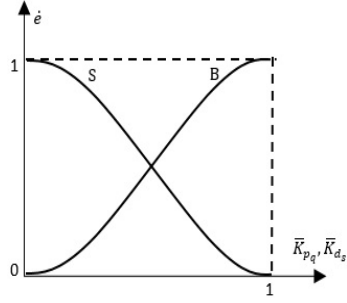


Figure 4: Membership functions of fuzzy logical outputs

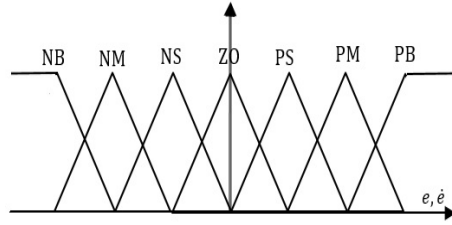


Figure 5: Membership functions of fuzzy logical inputs

Table 1: Rules for $\bar{K}_{pq}$

$\dot{e}$		NB	NM	NS	ZO	PS	PM	PB
e	NB	B	B	B	B	B	B	B
	NM	S	B	B	B	B	B	S
	NS	S	S	B	B	B	S	S
	ZO	S	S	S	B	S	S	S
	PS	S	S	B	B	B	S	S
	PM	S	B	B	B	B	B	S
	PB	B	B	B	B	B	B	B

Table 2: Rules for $\bar{K}_{ds}$

$\dot{e}$		NB	NM	NS	ZO	PS	PM	PB
e	NB	S	S	S	S	S	S	S
	NM	B	B	S	S	S	B	B
	NS	B	B	B	S	B	B	B
	ZO	B	B	B	B	B	B	B
	PS	B	B	B	S	B	B	B
	PM	B	B	S	S	S	B	B
	PB	S	S	S	S	S	S	S

$\mathcal{V}$ as

$$\begin{aligned}
\mathcal{V} = & \frac{1}{2}(r_1 v_y^2 + r_2 v_z^2 + r_3 \vartheta_1^2 + r_4 \Omega_1^2 + r_5 \vartheta_2^2 + r_6 \Omega_2^2 + r_7 \vartheta_3^2 + r_8 \Omega_3^2) \\
& + \frac{1}{2} \sum_{i=1}^2 \left[r_9 (\dot{\varphi}_i^2 + \frac{1}{m_i l_i^2 \sin^2 \varphi_i + I_i} (2\dot{\varphi}_i m_i l_i h_i \Omega_3 \cos \varphi_i \sin \theta_i - 2\dot{\varphi}_i m_i l_i \right. \\
& h_i \Omega_2 \cos \varphi_i \cos \theta_i + 2\dot{\varphi}_i I_i \Omega_2 \cos \theta_i - 2\dot{\varphi}_i I_i \Omega_3 \sin \theta_i) + r_{10} (\dot{\theta}_i^2 + \frac{1}{I_i} (2\dot{\theta}_i m_i \\
& l_i h_i \Omega_3 \sin \varphi_i \cos \theta_i + 2\dot{\theta}_i m_i l_i h_i \Omega_2 \sin \varphi_i \sin \theta_i - 2\dot{\theta}_i I_i \Omega_1)) \left. \right]
\end{aligned} \tag{56}$$

where r_{1-10} are positive constants, which can be chosen such that the candidate Lyapunov function is positive definite. The time derivative of $\mathcal{V}$ can be computed as

$$\dot{\mathcal{V}} = A + B u_1 + C u_2 + \sum_{i=1}^3 [(f_{i1} u_3 + f_{i2} u_4 + f_{i3} u_5 + g_i) \Omega_i] \tag{57}$$

where A , B , C , f_{i1} , f_{i2} , f_{i3} and g_i are functions of dynamic variables of the SC-fuel system. Choosing the new control inputs as

$$u_1 = -w_1 B \tag{58}$$

$$u_2 = -w_2 C \tag{59}$$

$$\begin{bmatrix} u_3 \\ u_4 \\ u_5 \end{bmatrix} = - \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix}^{-1} \begin{bmatrix} g_1 + w_3 \Omega_1 \\ g_2 + w_4 \Omega_2 \\ g_3 + w_5 \Omega_3 \end{bmatrix} \tag{60}$$

yields

$$\dot{\mathcal{V}} = -w_1 B^2 - w_2 C^2 - w_3 \Omega_1^2 - w_4 \Omega_2^2 - w_5 \Omega_3^2 + A$$

which can be rendered negative semi-definite by choosing the positive constants w_{1-5} large enough.

4. Numerical simulations

Simulations are conducted to illustrate the effectiveness of the controllers designed in the previous sections. For this purpose, a SC maneuvering in 3D space is considered. The sliding mode (43)-(47), fuzzy logic (49)-(53) and Lyapunov-based (58)-(60) controllers are employed to control the sloshing dynamics, transverse velocity vectors and attitude angles of the SC. Table 3 and Table 4 present the physical parameters of the SC and pendulums, and initial conditions. In

Table 3: Physical parameters of the system

Parameter	Value	Unit	Parameter	Value	Unit
m	975	Kg	l_1	0.2	m
j_x	600	$Kg.m^2$	l_2	0.1	m
j_y	800	$Kg.m^2$	h_0	0.05	m
j_z	400	$Kg.m^2$	h_1	0.6	m
m_0	480	Kg	h_2	0.9	m
I_0	75	$Kg.m^2$	b	-0.6	m
m_1	50	Kg	d	1.2	m
m_2	5	Kg	F	2450	N
I_1	10	$Kg.m^2$	ϵ_1	3.7	$Kg.m^2/s$
I_2	1	$Kg.m^2$	ϵ_2	0.5	$Kg.m^2/s$

addition, Table 5, Table 6 and Table 7 present the control parameters of the sliding mode, fuzzy logic and Lyapunov-based controllers, respectively.

By applying the designed controllers on reduced equations of motion, the effectiveness of the controllers on controlling the velocity vector and attitude

Table 4: Initial condition

Variable	Value	Unit	Variable	Value	Unit
v_{x_0}	3000	m/s	φ_{1_0}	30	<i>Degree</i>
v_{y_0}	75	m/s	φ_{2_0}	40	<i>Degree</i>
v_{z_0}	25	m/s	θ_{1_0}	20	<i>Degree</i>
ϑ_{1_0}	5	<i>Degree</i>	θ_{2_0}	10	<i>Degree</i>
ϑ_{2_0}	2	<i>Degree</i>	$\dot{\vartheta}_{1_0} = \dot{\vartheta}_{2_0} = \dot{\vartheta}_{3_0}$	0	rad/s
ϑ_{3_0}	-5	<i>Degree</i>	$\dot{\varphi}_{1_0} = \dot{\varphi}_{2_0} = \dot{\theta}_{1_0} = \dot{\theta}_{2_0}$	0	rad/s

Table 5: Control parameters of sliding mode controller

Control parameter	Value	Cotrol parameter	Value
$K_1 = K_2$	1×10^{-5}	$K_3 = K_4 = K_5$	5×10^{-5}
$\lambda_1 = \lambda_2$	2×10^{-2}	$\lambda_3 = \lambda_4 = \lambda_5$	2×10^{-2}
$B_1 = B_2$	2	$B_3 = B_4 = B_5$	0.2

Table 6: Maximum and minimum of fuzzy controller parameters

Maximum	Value	Minimum	Value
$K_{p1_{max}}$	$1.2 \times 4 \times 10^{-3}$	$K_{p1_{min}}$	$0.64 \times 4 \times 10^{-3}$
$K_{p2_{max}}$	$1.2 \times 4 \times 10^{-3}$	$K_{p2_{min}}$	$0.64 \times 4 \times 10^{-3}$
$K_{p3_{max}}$	8×10^{-5}	$K_{p3_{min}}$	$0.53 \times 8 \times 10^{-5}$
$K_{d3_{max}}$	$2 \times 5 \times 10^{-3}$	$K_{d3_{min}}$	5×10^{-3}
$K_{p4_{max}}$	8×10^{-5}	$K_{p4_{min}}$	$0.53 \times 8 \times 10^{-5}$
$K_{d4_{max}}$	$2 \times 5 \times 10^{-3}$	$K_{d4_{min}}$	5×10^{-3}
$K_{p5_{max}}$	8×10^{-5}	$K_{p5_{min}}$	$0.53 \times 8 \times 10^{-5}$
$K_{d5_{max}}$	$2 \times 5 \times 10^{-3}$	$K_{d5_{min}}$	5×10^{-3}

Table 7: Control parameters of Lyapunov-based controller

Control parameter	Value	Cotrol parameter	Value
$r_1 = r_2$	8×10^{-7}	$r_3 = r_5 = r_7$	1×10^3
$r_4 = r_6 = r_8$	500	$r_9 = r_{10}$	1×10^{-5}
$w_1 = w_2$	1×10^4	$w_3 = w_4 = w_5$	4×10^4

angles of the SC, and attitude angles of the pendulums, could be shown. Furthermore, the time behaviors of control inputs (M_x , M_y , M_z , δ_1 and δ_2) could be studied by considering (13)-(17).

First, the effectiveness of the sliding mode controller is studied. Figs. 6-11 show the simulation results for the sliding mode controller case. As Figs. 6, 7 indicate, the velocity vector and attitude angles of the SC are stabilized in around 200 seconds. As can be seen in Figs. 8, 9, the attitude angles of the pendulums are stabilized in around 400 seconds. Figs. 10, 11 show the time behaviors of the control inputs.

Next, the effectiveness of the fuzzy logic controller is studied. Figures 12-17 show the simulation results for the fuzzy logic controller case. As Figs. 12, 13 indicate, the velocity vector and attitude angles of the SC are stabilized in around 800 seconds. As can be seen in Figs. 14, 15, the attitude angles of the pendulums are stabilized in around 800 seconds as well. Figs. 16, 17 show the time behaviors of the control inputs.

Finally, Figs. 18-23 show the effectiveness of Lyapunov-based controller. As Figs. 18, 19 indicate, the velocity vector and attitude angles of the SC are stabilized in around 400 seconds. As can be seen in Figs. 20, 21, the attitude angles of the pendulums are stabilized in around 600 seconds. Figs. 22, 23 show the time behaviors of the control inputs.

We compare the performance of the designed controllers in terms of the stabilization (or settling) time in Figs. 24-29. As seen in Figs. 24-25 the sliding mode controller stabilizes the velocity of the SC in the least time, whereas, the attitude angles of the SC are stabilized in the least time by Lyapunov-based

controller. Moreover, the overshoot of the fuzzy controller in time behaviors of the SC variables is more than that of the other controllers, whereas overshoot of this controller in time behaviors of attitude angles of the pendulum and control inputs is less than that of the other controllers.

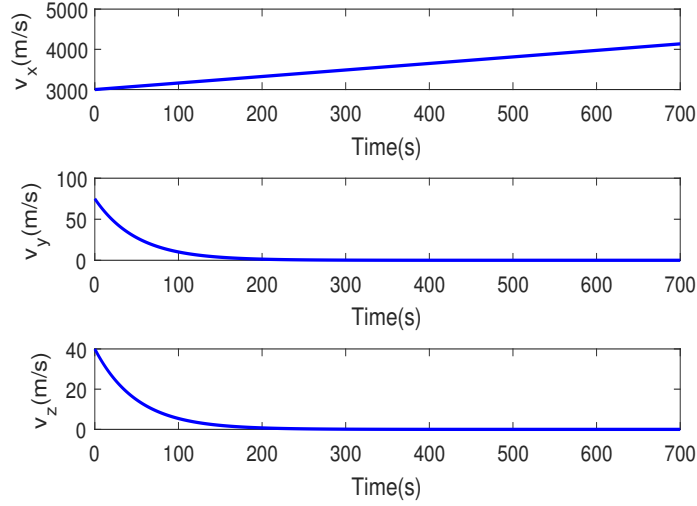


Figure 6: SC velocity (sliding mode controller)

5. Conclusion

The sloshing phenomenon in a partially filled fuel tank has been studied. Fuel sloshing is described by 3D two-pendulum model to characterize the first two prominent sloshing modes. Coupled Equations of the SC-fuel system have been derived using a quasi-Lagrangian formulation. Sliding mode, fuzzy logic and Lyapunov-based controllers are designed to control the attitude of the SC while suppressing the fuel slosh. Results of simulations are included to illustrate the effectiveness of the controllers designed in the paper. As concluded from the comparison results with respect to the stabilization times, sliding mode controller has better performance in the stabilization of SC velocity, whereas

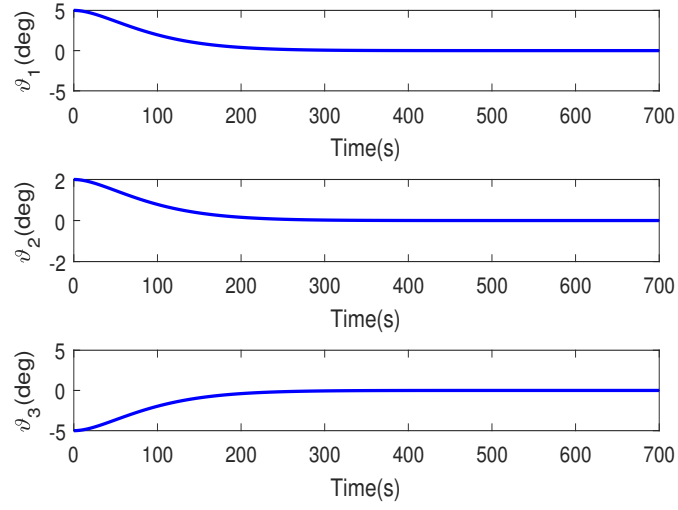


Figure 7: SC attitude (sliding mode controller)

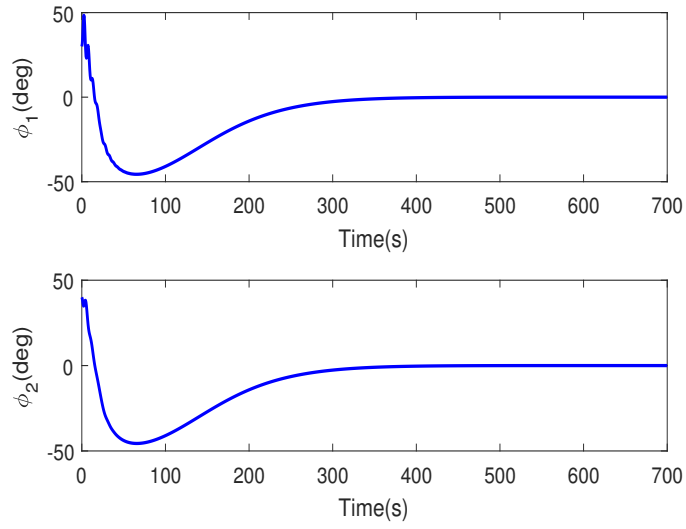


Figure 8: Time responses of φ_i (sliding mode controller)

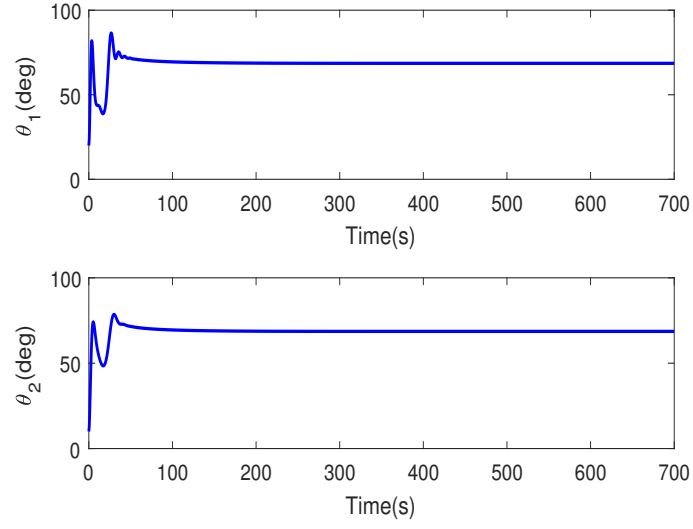


Figure 9: Time responses of θ_i (sliding mode controller)

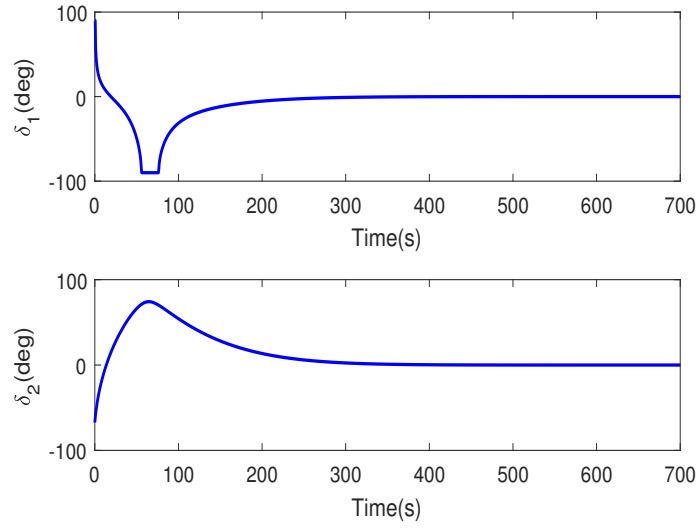


Figure 10: Time responses of $\delta_{1,2}$ (sliding mode controller)

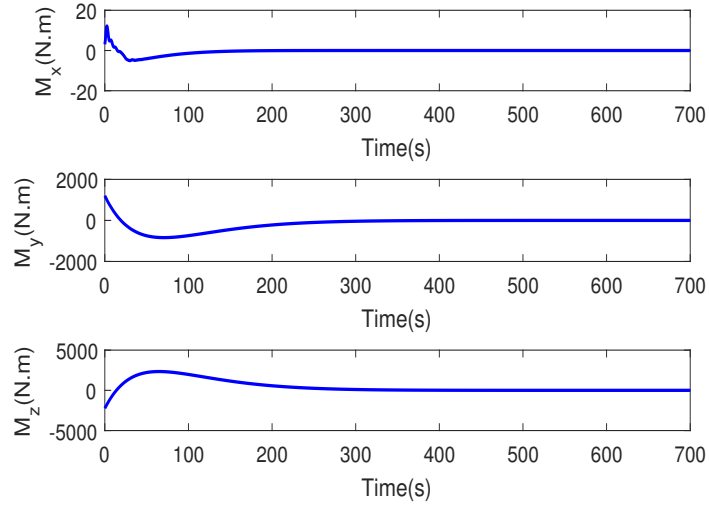


Figure 11: Time responses of $M_{x,y,z}$ (sliding mode controller)

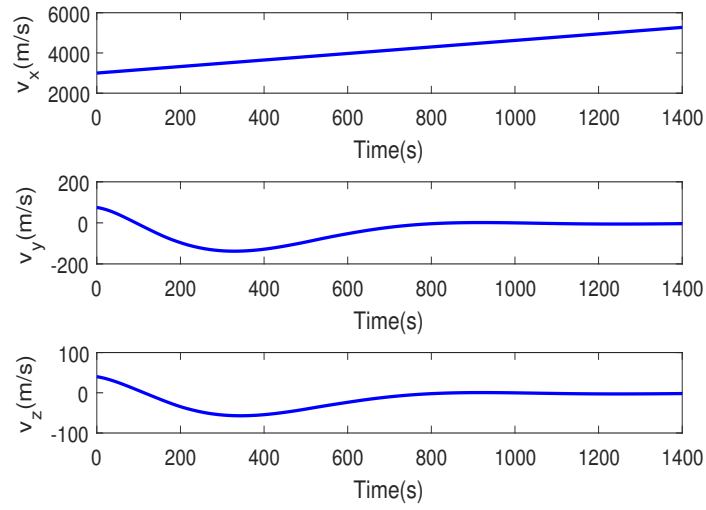


Figure 12: SC velocity (fuzzy controller)

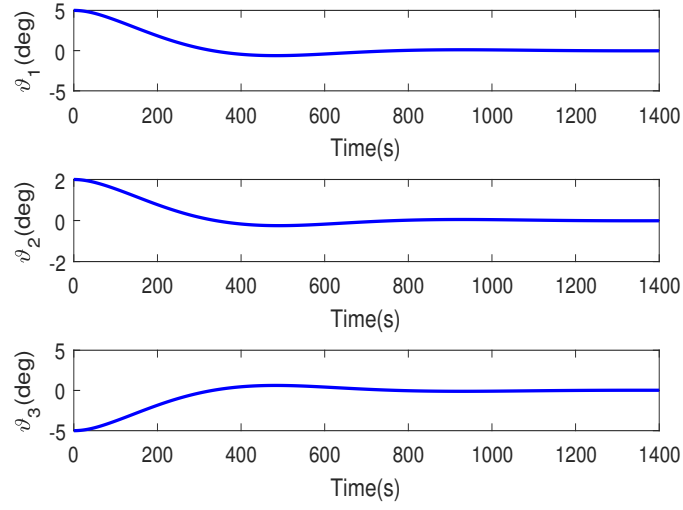


Figure 13: SC attitude (fuzzy controller)

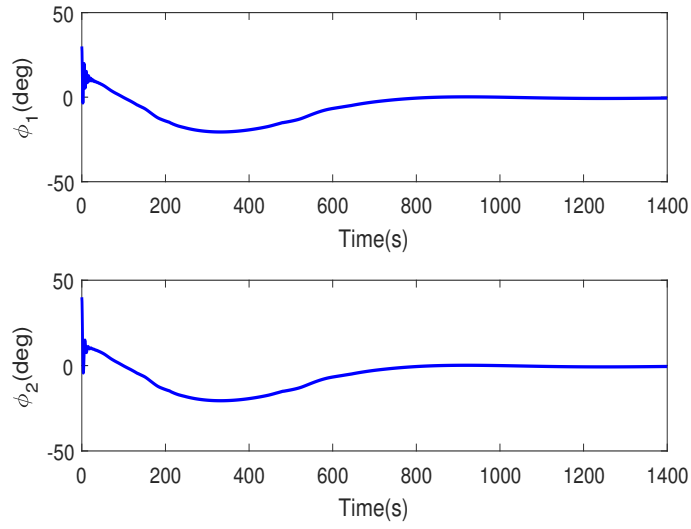


Figure 14: Time responses of φ_i (fuzzy controller)

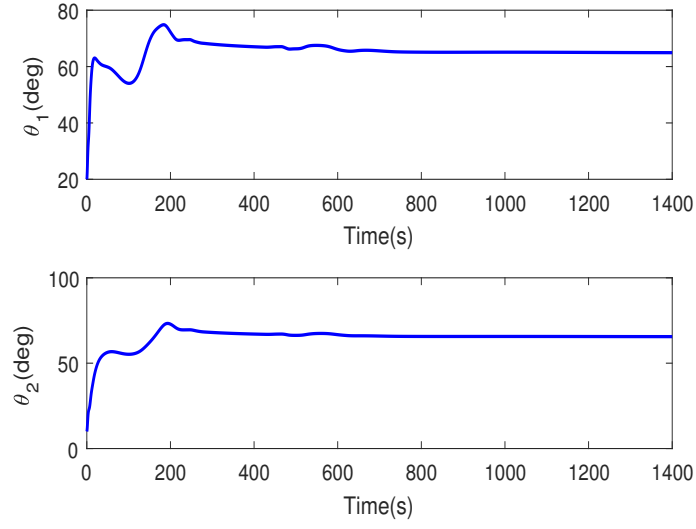


Figure 15: Time responses of θ_i (fuzzy controller)

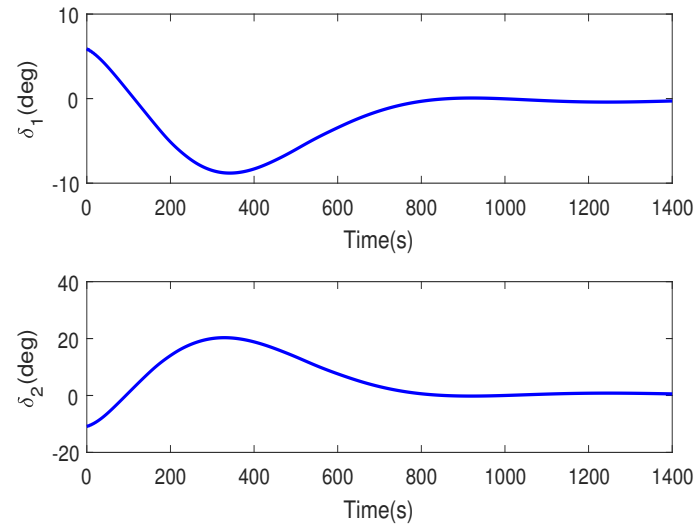


Figure 16: Time responses of $\delta_{1,2}$ (fuzzy controller)

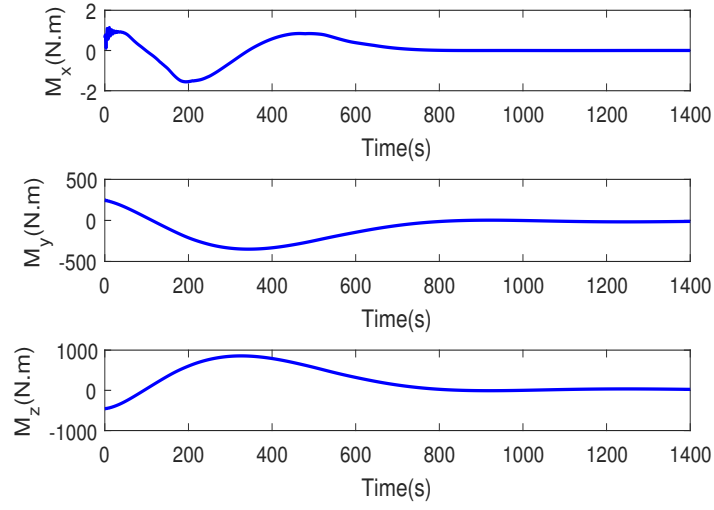


Figure 17: Time responses of $M_{x,y,z}$ (fuzzy controller)

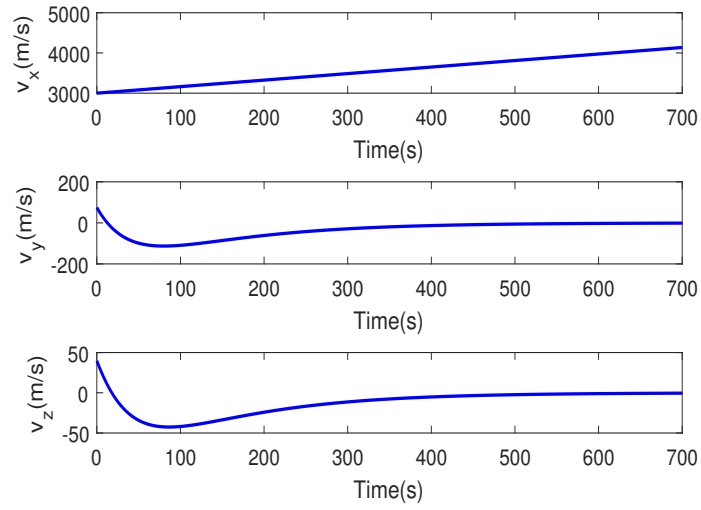


Figure 18: SC velocity (Lyapunov-based controller)

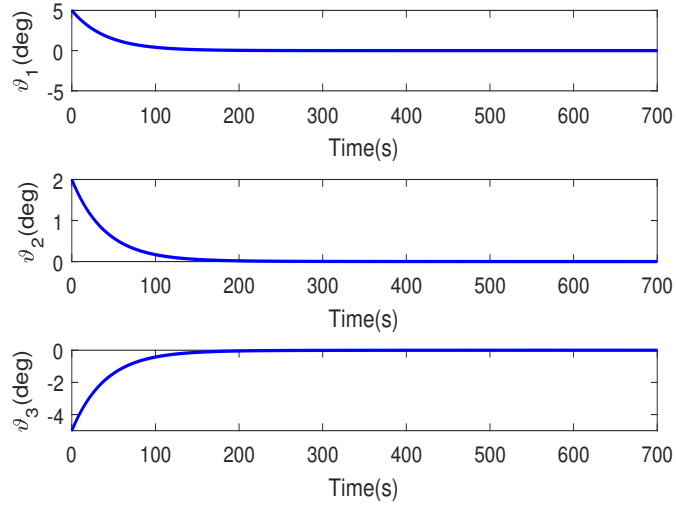


Figure 19: SC attitude angles (Lyapunov-based controller)

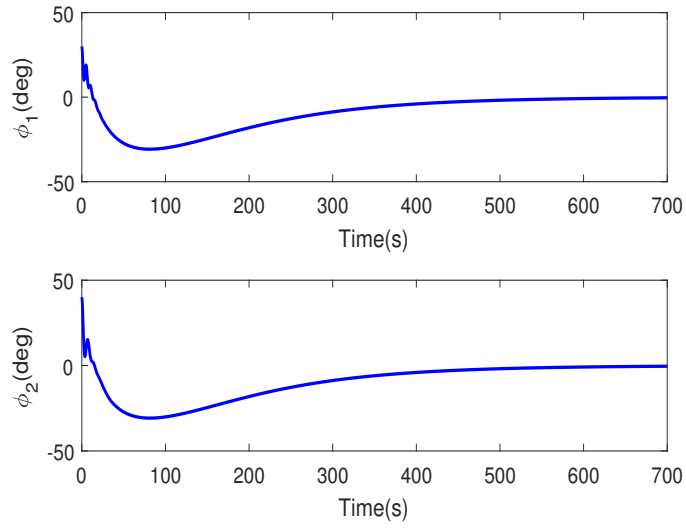


Figure 20: Time responses of φ_i (Lyapunov-based controller)

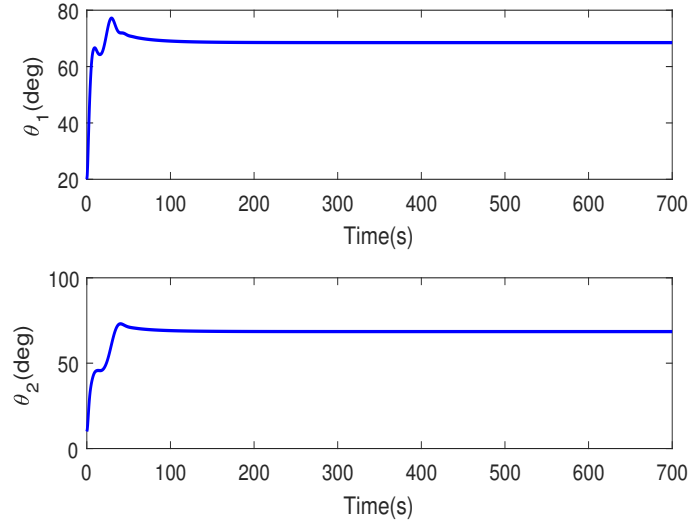


Figure 21: Time responses of θ_i (Lyapunov-based controller)

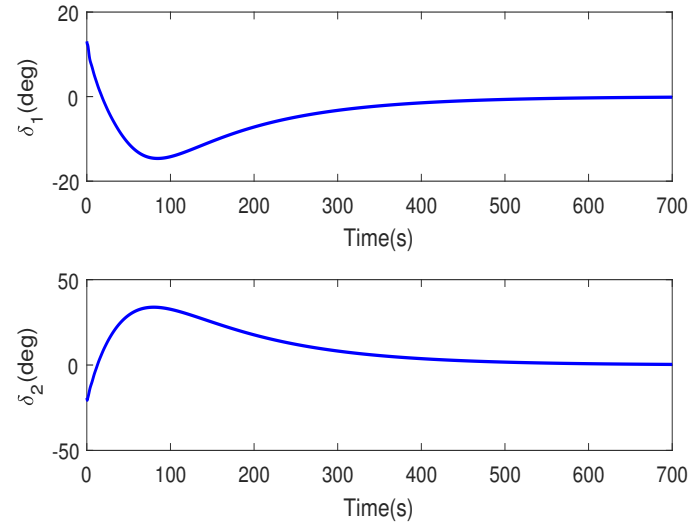


Figure 22: Time responses of $\delta_{1,2}$ (Lyapunov-based controller)

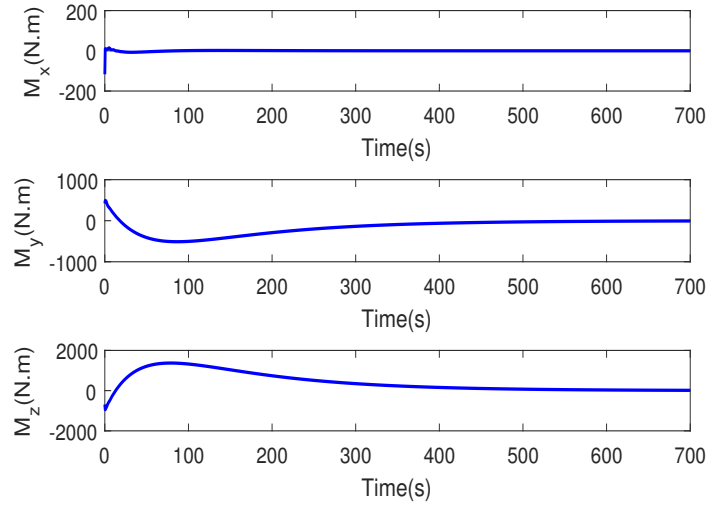


Figure 23: Time responses of $M_{x,y,z}$ (Lyapunov-based controller)

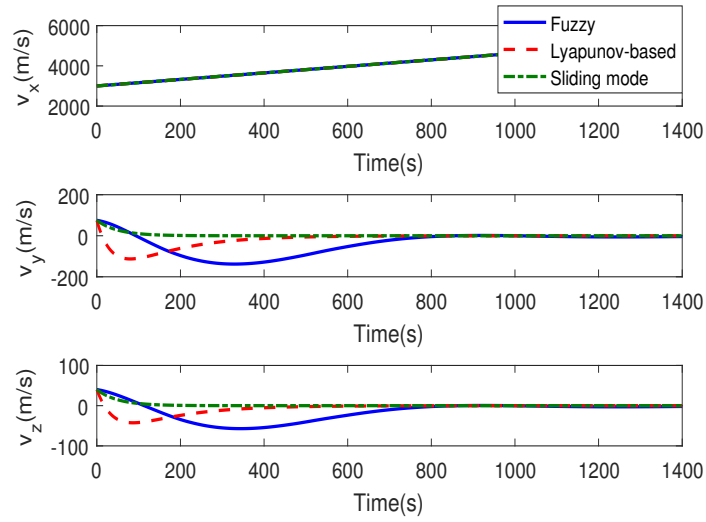


Figure 24: Comparison of time responses of SC velocity

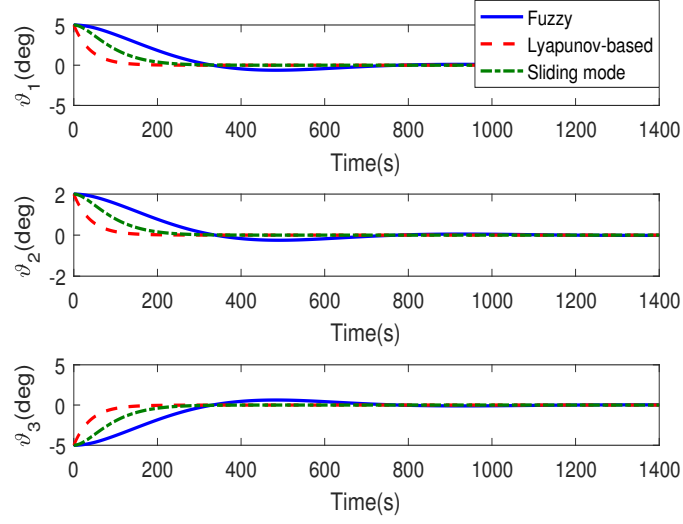


Figure 25: Comparison of time responses of SC attitude

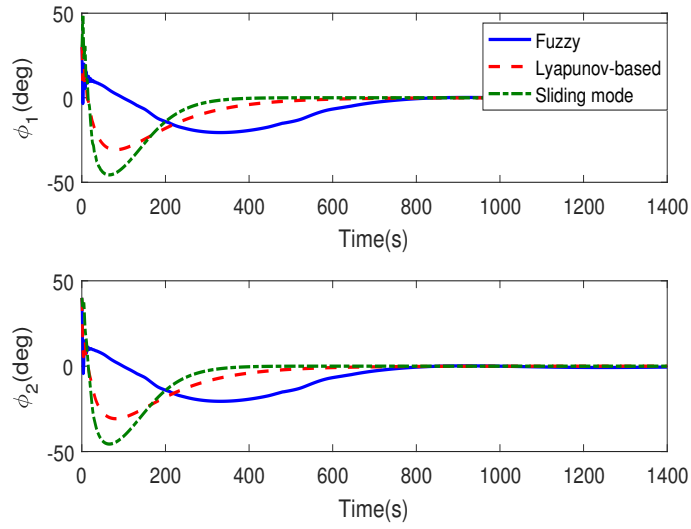


Figure 26: Comparison of time responses of φ_i

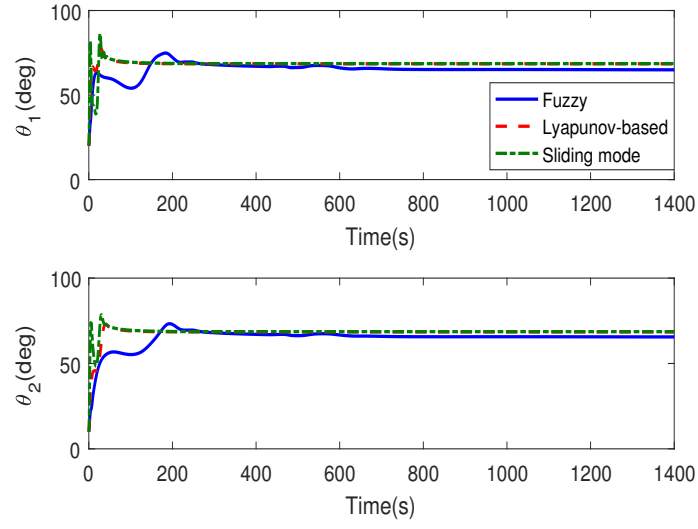


Figure 27: Comparison of time responses of θ_i

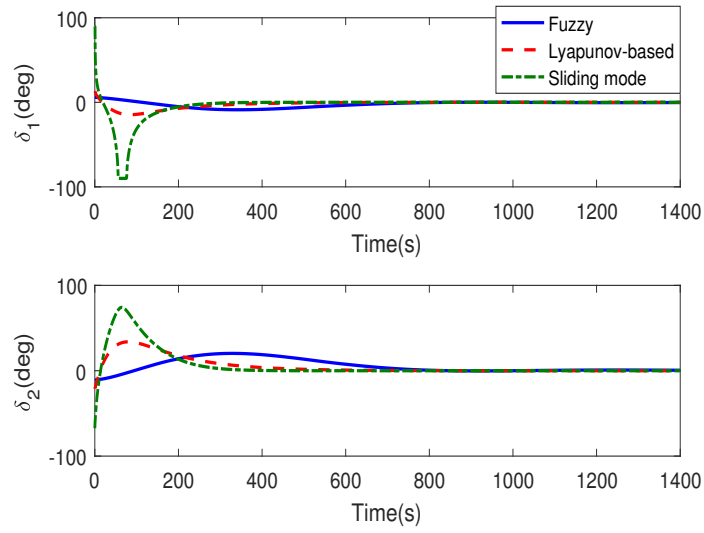


Figure 28: Comparison of time responses of $\delta_{1,2}$

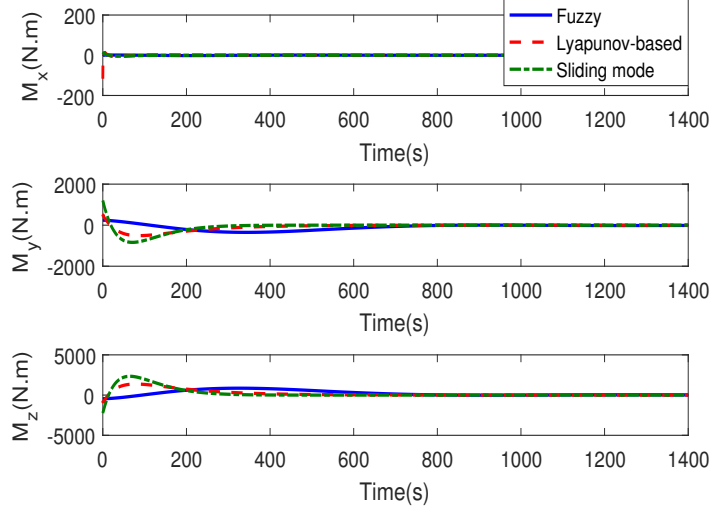


Figure 29: Comparison of time responses of $M_{x,y,z}$

Lyapunov-based controller has better performance in the stabilization of SC attitude. Moreover, the stabilization time of state variables using fuzzy logic controller takes more time than that of other controllers.

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